

Fall 2025 CPTS530 Homework 4  
MATH 448-548  
Numerical Analysis

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## Problem 1

**Problem Statement (Textbook 4.1.7):**

Let  $A$  have the block form

$$A = \begin{bmatrix} B & C \\ 0 & I \end{bmatrix}$$

in which the blocks are  $n \times n$ . Prove that if  $(B - I)$  is nonsingular, then, for  $k \geq 1$ ,

$$A^k = \begin{bmatrix} B^k & (B^k - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix}$$

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**Solution:**

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## Problem 2

**Problem Statement (Textbook 4.1.12):**

Let  $A$  be an  $n \times n$  invertible matrix, and let  $u$  and  $v$  be two vectors in  $\mathbb{R}^n$ . Find the necessary and sufficient conditions on  $u$  and  $v$  in order that the matrix

$$\begin{bmatrix} A & u \\ v^T & 0 \end{bmatrix}$$

be invertible, and give a formula for the inverse when it exists.

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**Solution:**

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## Problem 3

### Problem Statement (Textbook 4.1.17):

A square matrix  $A$  is said to be **skew-symmetric** if  $A^T = -A$ . Prove that if  $A$  is skew-symmetric, then  $x^T Ax = 0$  for all  $x$ .

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**Solution:**

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## Problem 4

**Problem Statement (Textbook 4.1.18):**

(Continuation) Prove that the diagonal elements of a skew-symmetric matrix are **0**. Also, prove that the determinant is 0 when the matrix is of odd order.

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**Solution:**

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## Problem 5

**Problem Statement (Textbook 4.1.19):**

(Continuation) Let  $A$  be any square matrix, and define  $A_0 = \frac{1}{2}(A + A^T)$  and  $A_1 = \frac{1}{2}(A - A^T)$ . Prove that  $A_0$  is symmetric, that  $A_1$  is skew-symmetric, that  $A = A_0 + A_1$  and that for all  $x$ ,  $x^T A x = x^T A_0 x$ . This explains why, in discussing quadratic forms, we can confine our attention to **symmetric matrices**.

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**Solution:**

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## Problem 6

### Problem Statement (Textbook 4.2):

This problem has multiple parts related to LU and Cholesky factorizations.

### Part 1 (Textbook 4.2.1)

Prove these facts, needed in the proof of Theorem 2.

- (a) If  $U$  is upper triangular and invertible, then  $U^{-1}$  is upper triangular.
- (b) The inverse of a unit lower triangular matrix is unit lower triangular.
- (c) The product of two upper (lower) triangular matrices is upper (lower) triangular.

### Part 5 (Textbook 4.2.5)

Prove that an upper or lower triangular matrix is nonsingular if and only if its diagonal elements are all different from 0.

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**Solution:**

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## Appendix

This appendix contains the complete codebase for the problems solved in this homework. All implementations are written in Julia and developed from scratch without using built-in numerical solver shortcuts.

**Problem 1: Code and Figures**

**Problem 2: Code and Figures**

**Problem 3: Code and Figures**

**Problem 4: Code and Figures**

**Problem 5: Code and Figures**

**Problem 6: Code and Figures**